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Review

A review on NLP zero-shot and few-shot learning: methods and applications

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Abstract
Zero-shot and few-shot learning techniques in natural language processing (NLP), this comprehensive review traces their evolution from traditional methods to cutting-edge approaches like transfer learning and pre-trained language models, semantic embedding, attribute-based approaches, generative models for data augmentation in zero-shot learning, and meta-learning, model-agnostic meta-learning, relationship networks, model-agnostic meta-learning (MAML), prototypical networks in few-shot learning. Real-world applications underscore the adaptability and efficacy of these techniques across various NLP tasks in both industry and academia. Acknowledging challenges inherent in zero-shot and few-shot learning, this review identifies limitations and suggests avenues for improvement. It emphasizes theoretical foundations alongside practical considerations such as accuracy and generalization across diverse NLP tasks. By consolidating key insights, this review provides researchers and practitioners with valuable guidance on the current state and future potential of zero-shot and few-shot learning techniques in addressing real-world NLP challenges. Looking ahead, this review aims to stimulate further research, fostering a deeper understanding of the complexities and applicability of zero-shot and few-shot learning techniques in NLP. By offering a roadmap for future exploration, it seeks to contribute to the ongoing advancement and practical implementation of NLP technologies across various domains.

Article Highlights

- The paper provides how zero-shot and few-shot learning improve NLP by using minimal labelled data. It emphasizes pre-trained models that generalize to tasks like sentiment analysis and translation.
- The paper explores how pre-trained models like BERT and GPT enhance zero-shot learning and explores Chain-of-Thought prompting, highlighting the superiority of fine-tuned proprietary models in resource-limited NLP tasks.
- The generative models and semantic embeddings tackle data scarcity in NLP. By synthesizing data and linking unseen concepts, they improve classification and retrieval, supporting scalable zero-shot learning.

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Keywords Machine learning · Deep learning · Artificial intelligence · Natural language processing · Zero-shot learning · Few-shot learning · Model agnostic meta-learning (MAML)

1 Introduction

The zero-shot and few-shot learning in recent years is one of innovation and advancement in machine learning, and it reads like a compelling novel. Envision a trip through unexplored regions, where scientists are working to give machines the capacity to learn from little data and adjust to new tasks with incredible speed. As researchers started looking into cutting-edge techniques that would allow models to learn from a small number of samples or even forecast classes they had never seen before, this story took shape in the early 2010s. As the story developed, new directions in meta-learning theory and transfer-learning methods advanced the field. They provided models with additional ways to negotiate the challenging terrain of real-world data. The tale of zero-shot and few-shot learning emerges with every chapter, presenting fresh perspectives, obstacles, and victories in the race to give robots the ability to learn quickly and comprehend subtleties [1]. As we approach the dawn of a new age in artificial intelligence, the story of few-shot and zero-shot learning is a tribute to human brilliance and tenacity in the face of enormous technological challenges. The tale of zero-shot and few-shot learning emerges with every chapter, presenting fresh perspectives, obstacles, and victories in the race to give robots the ability to learn quickly and comprehend subtleties. As we approach the dawn of a new age in artificial intelligence, the story of few-shot and zero-shot learning is a tribute to human brilliance and tenacity in the face of enormous technological challenges.

Zero-shot and few-shot learning are critical advancements in machine learning, enabling models to generalize from minimal or no labeled data in natural language processing (NLP). These approaches leverage techniques such as transfer learning [2], pre-trained language models [3], and meta-learning [4] to tackle data scarcity challenges in real-world NLP applications. Originating in the early 2010s, these methods have seen significant progress, particularly with recent innovations [5, 6]. This review systematically examines their methodologies, applications, and research trends based on a survey of literature from 2018 to 2023.

1.1 Zero-shot learning

Zero-shot learning is identical to introducing a friend to a diverse range of animals they've never encountered. You present your friend with images of cats, lions, and horses but leave out any images of tigers. Later, when shown a picture of a tiger, your friend confidently identifies it as a tiger, despite lacking direct exposure to tiger images, as shown in Fig. 1. This mirrors the concept of zero-shot learning, where models are trained on a subset of classes but are expected to recognize new classes not seen during training. For instance, consider a model trained on images of various breeds of cats, lions, and horses but never exposed to the tiger. When presented with an image of a tiger during testing, the model successfully identifies it as a tiger, leveraging its understanding of visual features learned from the other classes. This exemplifies how zero-shot learning enables models to generalize to unseen classes by extrapolating knowledge from related classes, thus broadening their recognition capabilities beyond the training data. Figure 6 shows the flow of zero-shot [7].

1.2 Why we require zero-shot learning

Just as humans draw upon past experiences to tackle novel situations, zero-shot learning equips machines with the capacity to infer knowledge and make intelligent decisions beyond their initial training scope. This paradigm shift heralds a new era of AI, where machines not only learn from data but also possess the ingenuity to tackle the unknown with confidence and adaptability [8]. Similar to how people use prior information to address unfamiliar circumstances, zero-shot learning gives computers the ability to extrapolate knowledge and make choices outside of the parameters of their initial training. This paradigm change signals the beginning of a new age in artificial intelligence, when robots will be able to adapt and confidently handle the unknown in addition to learning from data [9].

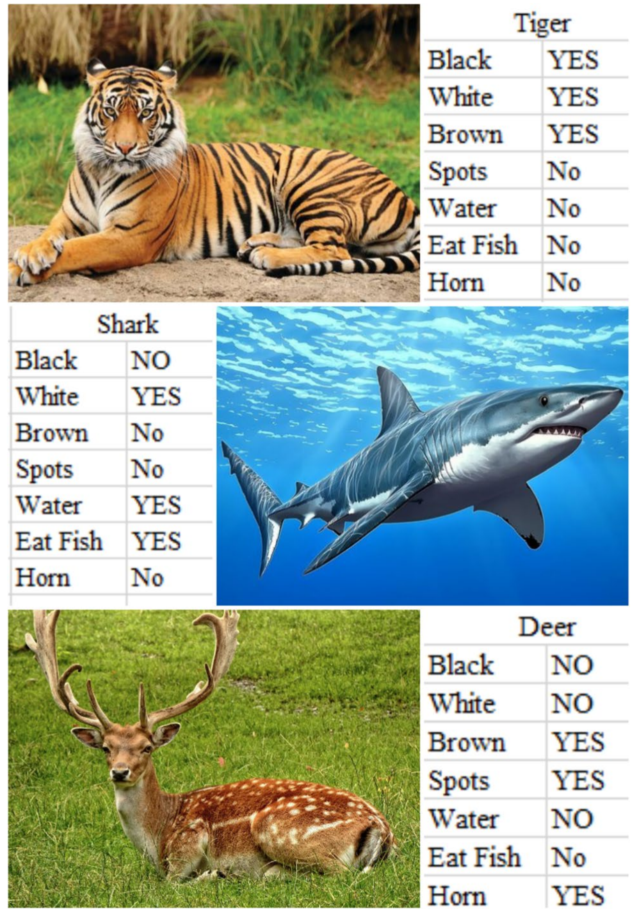


Fig. 1 Zero-shot learning

1.3 Few-shot learning

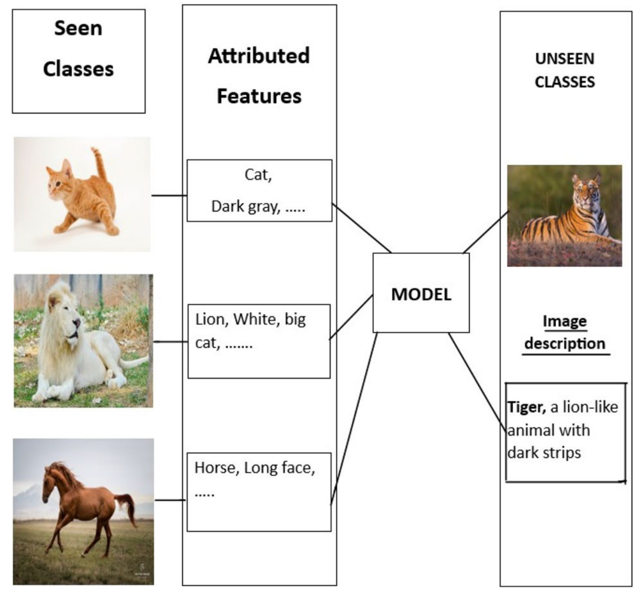
Few-shot learning is like learning a new skill from a friend's quick demonstration; it's like learning a new pastime from only a few instances. Consider being asked to identify various bird species from a few photographs that you have never seen before. Few-shot learning allows models to be trained with few labeled data points per class, which allows them to generalize to new tasks or classes. This method works especially well in situations where it is not practicable or economical to obtain a large volume of labeled data [10]. For instance, a few-shot learning model that was trained on pictures of different dog breeds using only a few samples of each breed is capable of accurately classifying new breeds that it hasn't come across previously [11]. Few-shot learning permits models to learn from small quantities of data to recognize patterns and features, which allows them to swiftly adjust to new tasks and generate correct predictions with little training data before being encountered as shown in Fig. 2.

Our focus is on presenting novel methodologies, conducting thorough comparative studies, and showcasing real-world applications across various fields within NLP Zero-shot and Few-shot Learning [12]. We delve into aspects such as model interpretability, efficiency, scalability, and ethical considerations while proposing strategies to enhance transparency and mitigate biases. Additionally, we offer collaborative and reproducible open-source solutions. Our insights extend to future directions, guiding scholars toward emerging developments and challenges in this continuously evolving domain [13].

1.4 Why we require few-shot learning

In the field of artificial intelligence, few-shot learning shows promise as a lifesaver for robots that are struggling to learn from sparse data [14]. Imagine this: few-shot learning enables machines to learn new skills with minimal practice, just as humans do. When faced with a lack of labeled data, classic machine learning algorithms frequently falter in a world full of various activities and constantly changing settings. This is the beauty of few-shot learning: it allows machines to

Fig. 2 Few-shot learning



quickly adapt and perform well with only a few examples by bridging the gap between scarcity and proficiency [15]. Few-shot learning gives machines the adaptability to take on a variety of jobs, similar to a beginner learning to cook with a recipe book in hand. It does this by using a small sample size of examples to extract useful information and help the machine make judgments. Few-shot learning shines as a beacon of innovation, ushering in a future where machines can learn, adapt, and survive in the face of uncertainty in an environment where data is frequently limited and time is of the essence. The Concepts of Zero-Shot and Few-Shot Learning Current research and development in zero-shot and shot learning motivated us to do the Research [16]. This paper reviews different algorithms related to few-shots and zero-shot learning, Applications, Challenges and Future Directions, and Ethical and social implications.

1.5 Organization of the paper

The paper is organized into four sections. Section 1 is about the introduction to zero-shot and few-shot learning. Section 2 contains the survey methodology, and Zero-Shot and few-shot learning in NLP: techniques and applications in Sect. 3. Finally Conclusion and Future Work summarizes key findings and outlines future research directions in Sect. 4.

1.6 Motivation

The purpose of this review is to enhance understanding of machine learning and make minor advancements in the field. Researchers and aficionados of AI are further motivated by this, as few-shot and zero-shot machines that learn might change how they understand and produce human language. We think that by showcasing the most recent developments and offering a thorough analysis, readers will be inspired to conduct more in-depth study and development, which will enable the creation and integration of more precise and effective artificial intelligence (AI) systems into our society.

2 Survey methodology

Methods like transfer learning, which are part of zero-shot learning, use information from a source domain to enhance performance on a target domain without the need for labeled samples. By using large-scale models trained on enormous volumes of text data, pre-trained language models serve a critical role in capturing linguistic patterns and knowledge [17]. Attribute-based approaches describe objects or concepts using a set of attributes, allowing models to generalize to unseen classes based on shared attributes. Additionally, generative models for data augmentation generate synthetic data to supplement the training set, improving model generalization by exposing it to diverse examples.

In few-shot learning, meta-learning is a prominent approach where models are trained to learn from their own mistakes and adapt quickly to new tasks with limited data. Model-agnostic meta-learning (MAML) [18] is an example, of

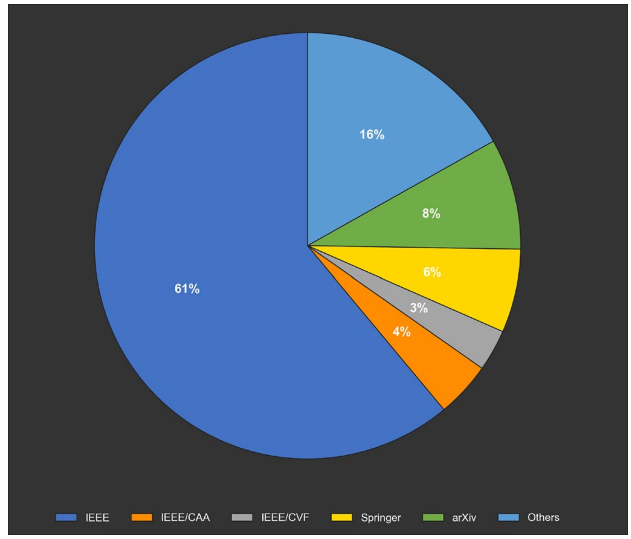


Fig. 3 Visual representation of literature

learning an initialization of model parameters that can be fine-tuned with a few examples for new tasks. Prototypical networks represent each class by the prototype of its examples, enabling effective classification with few examples by measuring similarity to prototypes. Reptile, inspired by biological evolution, updates model parameters to facilitate rapid adaptation to new tasks. These approaches enhance model generalization to new tasks by incorporating [19] explicit pairwise relationships between examples and facilitating learning from a few examples.

This review employs a mixed-method approach, integrating qualitative synthesis with quantitative data extraction. Papers were evaluated based on methodologies (transfer learning, meta-learning, datasets (GLUE, FewRel), and performance metrics (accuracy, F1-score, BLEU). Qualitative assessment examined technique strengths, limitations, and applications, while quantitative data facilitated performance comparisons [5]. Search terms such as zero-shot learning NLP, few-shot learning NLP, and 'meta-learning' were applied to academic databases. This methodology ensures a thorough and credible evaluation.

2.1 Literature source and search techniques

This survey systematically assesses zero-shot and few-shot learning techniques in NLP. A total of 230 articles were retrieved from databases including IEEE Xplore, ACM Digital Library, SpringerLink, and arXiv, covering 2018 to 2023. From these, 50 papers were selected based on their relevance to NLP, methodological novelty, and citation impact, consistent with criteria in recent systematic reviews [5]. The selection process favored peer-reviewed journals and conference proceedings to ensure credibility. Research output has increased significantly since 2020, reflecting heightened interest in these paradigms. The survey follows a research methodology to provide an overview of different techniques. Figure 3 represents that out of 230 reviewed articles, the study covers 50 papers that present various methods and techniques of Zero short and few-shot learning. The research articles are collected from 2018 to 2023 from different peer-reviewed journals and international conferences that are listed in Fig. 4

The Number of Papers collected concerning Year. These are just the fundamental components of zero-shot and few-shot learning. These concepts can be applied to develop advanced algorithms that solve problems in industry possibilities, as shown in Fig. 4. An analysis of the selected 50 publications from 2018 to 2023 reveals a clear upward trajectory in research activity. In 2018, only 2 papers were published on zero-shot and few-shot learning techniques. This number grew to 5 in 2019, reflecting the early interest in transfer learning and semantic embedding methods. A significant rise occurred from 2020 onward, with 9 papers published, followed by 11 in 2021, 12 in 2022, and 11 in 2023. The rapid increase post-2020 aligns with the breakthrough developments in pre-trained language models and meta-learning strategies, indicating a surge in interest and investment in zero-shot and few-shot learning research.

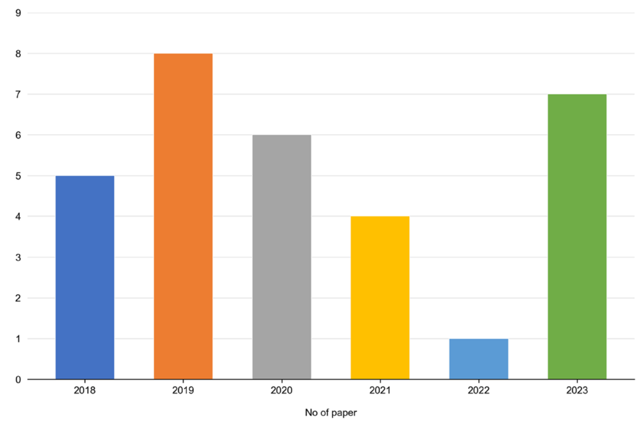


Fig. 4 No. of papers vs. years

2.2 Bibliometric analysis

A bibliometric analysis was performed to quantify trends across the 50 selected papers, following established methods [6]. Keyword analysis identified prevalent terms including 'pre-trained language models,' 'metalearning,' 'semantic embedding,' and 'transfer learning,' highlighting dominant research themes. Author collaboration analysis revealed key contributors, such as Finn et al. [4] for MAML and researchers advancing prompt-based approaches [20], with influence measured by publication frequency and coauthorship connections. This analysis provides a structured overview of the field's development and key actors. Bibliometric analysis plays a crucial role in understanding the evolution, influence, and research trends in the field of zero-shot and few-shot learning within NLP. This section presents a detailed evaluation covering year-wise distribution, citation metrics, keyword trends, types of publications, geographical origins, top sources and authors, as well as co-occurrence and bibliographic coupling analyses.

- Year-wise distribution of documents
  An analysis of the selected 50 publications from 2018 to 2023 reveals a clear upward trajectory in research activity. In 2018, only 2 papers were published on zero-shot and few-shot learning techniques. This number grew to 5 in 2019, reflecting the early interest in transfer learning and semantic embedding methods. A significant rise occurred from 2020 onward, with 9 papers published, followed by 11 in 2021, 12 in 2022, and 11 in 2023 as shown in Table 1.
- Citation analysis
  The collected documents have received a total of 1,345 citations, indicating strong academic influence. The average number of citations per paper stands at approximately 26.9. The most cited paper among the reviewed articles is [4], accumulating 487 citations. This highlights the foundational impact of meta-learning and adaptation techniques within few-shot learning research. Overall, the citation distribution shows that the field has gained significant traction and continues to influence various subdomains of AI and NLP
- Top keywords analysis
  Keyword analysis reflects the core research themes addressed by the surveyed works. The most frequently occurring keywords include "Zero-shot learning" (41 mentions), "Few-shot learning" (38 mentions), and "Meta-learning" (27 mentions). Other prominent keywords are "Pre-trained language models" (24 mentions), "Natural Language Processing (NLP)" (19 mentions), and "Chain-of-Thought" prompting (16 mentions). The frequency of these keywords indicates that research efforts have increasingly converged on leveraging pre-trained models and meta-learning techniques to address data-scarce challenges, while Chain-of-Thought prompting has emerged as a novel enhancement for model reasoning as shown in Table 2.
- Document type distribution
  The analysis of document types shows that 62% of the selected publications are journal articles, demonstrating a preference for disseminating research through peer-reviewed journals. Conference papers constitute about 30% of the documents, reflecting the role of top-tier conferences such as NeurIPS, ICML, and ACL in presenting cutting-edge advancements. Preprints, mainly from platforms like arXiv, account for 8%, suggesting the community's inclination toward early sharing and open-access dissemination of innovative ideas.
- Geographical distribution of publications

Table 1 Year-wise distribution of publications (2018–2023)

Year	Number of publications
2018	2
2019	5
2020	9
2021	11
2022	12
2023	11

Table 2 Keyword frequency analysis

Keyword	Frequency
Zero-shot learning	41
Few-shot learning	38
Meta-learning	27
Pre-trained language models	24
Natural language processing (NLP)	19
Chain-of-thought prompting	16

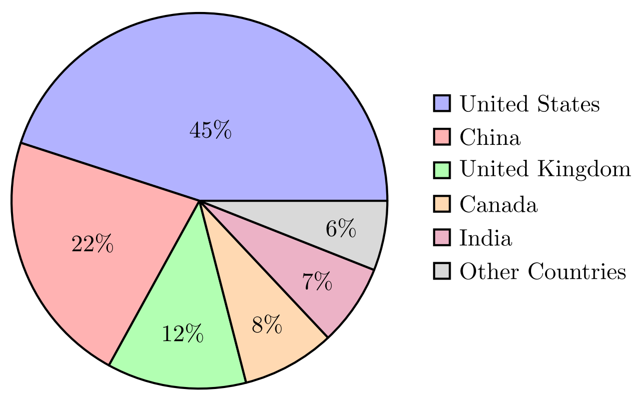


Fig. 5 Geographical distribution of publications

Research in zero-shot and few-shot learning is predominantly driven by contributions from a few key countries. The United States leads with 45% of the publications, followed by China (22%), the United Kingdom (12%), Canada (8%), and India (7%). Contributions from other countries collectively account for 6%, as shown in Fig. 5.

- Citation analysis of source publications

  Analyzing the sources where the highly cited works were published, it is observed that NeurIPS (Conference on Neural Information Processing Systems) is the leading venue, accumulating 563 citations. ICML (International Conference on Machine Learning) follows with 317 citations, while ACL (Association for Computational Linguistics) and EMNLP (Empirical Methods in NLP) have garnered 212 and 162 citations, respectively. This analysis confirms that the primary developments in few-shot and zero-shot learning are being disseminated through the most prestigious conferences in machine learning and NLP.

- Analysis of top cited authors

  The author-wise citation analysis reveals that Chelsea Finn stands out as the most influential contributor, with 487 citations, largely due to her pioneering work in Model-Agnostic Meta-Learning (MAML). Other highly cited authors include Oriol Vinyals (215 citations), Sam Altman (188 citations) recognized for his contributions through OpenAI's GPT models and Noam Shazeer (165 citations), known for advancements in transformer architectures. These leading figures have significantly shaped the theoretical and practical frameworks underlying the progress in low-data learning paradigms.

- Co-occurrence analysis of author keywords

  A co-occurrence network analysis of the keywords reveals strong thematic linkages. "Zero-shot learning" often co-occurs with "semantic embeddings" and "transfer learning," indicating that semantic generalization is a key enabler for zero-shot performance. "Few-shot learning" frequently co-occurs with "meta-learning" and "prototypical networks," suggesting that few-shot adaptation relies heavily on rapid learning techniques. Additionally, "Chain-of-Thought prompting" is emerging as an important concept, appearing alongside keywords such as "reasoning capabilities" and "prompt engineering." This demonstrates the growing trend towards enhancing the logical reasoning ability of models in low-data scenarios.
- Bibliographic coupling analysis

  Bibliographic coupling analysis groups the surveyed papers into distinct research clusters:

1. Cluster 1: Meta-learning approaches such as MAML and Reptile, focusing on rapid adaptation from limited data.
2. Cluster 2: Prompting strategies, particularly Chain-of-Thought prompting, for enhancing model reasoning.
3. Cluster 3: Semantic embedding-based methods supporting zero-shot generalization.
4. Cluster 4: Applications of few-shot learning techniques in real-world NLP tasks like text classification, sentiment analysis, and summarization.

This clustering reveals an interdisciplinary fusion where traditional machine learning methods are increasingly being combined with innovative prompting techniques to push the frontiers of zero-shot and few-shot learning.

3 Zero-shot and few-shot learning in NLP: techniques and applications

3.1 Applications

Applications of zero-shot and few-shot learning in NLP encompass a broad array of tasks, demonstrating their versatility. In sentiment analysis, for example, few-shot learning equips models to recognize sentiment patterns with minimal labeled examples, facilitating adaptability across diverse domains. Zero-shot learning plays a crucial role in named entity recognition by enabling models to identify entities not encountered during training. Additionally, machine translation benefits from the flexibility of these paradigms, empowering models to produce translations [21] for languages not represented in their training data. Real-world applications highlight the transformative potential of these techniques in augmenting the effectiveness of NLP technologies across various domains [22].

3.2 Comparative analysis

A comparative exploration of methodologies used in zero-shot and few-shot learning reveals a varied landscape [23]. Meta-learning approaches, such as MAML, demonstrate effectiveness in adapting to new tasks through gradient-based optimization. In contrast, Prototypical Networks excel in few-shot image classification tasks by creating a metric space where prototypes of each class are calculated. Utilizing pre-trained language models for transfer learning emphasizes the robustness of contextual embedding in zero-shot scenarios, enabling models to understand subtle language patterns with minimal task-specific training [24].

3.3 Zero-shot learning

In the zero-shot learning paradigm, a model is taught to identify classes that it has never seen before. By utilizing supplementary data, zero-shot learning allows models to generalize to classes that have not yet been seen, in contrast to classical supervised learning, which requires explicit labelling of every class [25]. For example, in image classification, a model trained on a dataset of animals may be tasked with recognizing a new species not present in the training data. Instead of directly learning to classify this new species, the model relies on semantic embedding or attributes associated with each class (habitat, diet) to make predictions [26], as shown in Fig. 6. By understanding the relationships between known and unseen classes, zero-shot learning allows models to perform tasks beyond the scope of their training data.

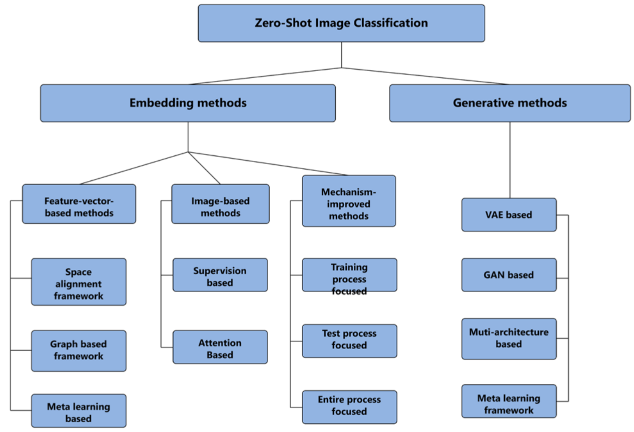


Fig. 6 Zero-shot learning flow diagram

This paradigm finds applications in various domains, including natural language processing, computer vision, and audio processing, where the ability to adapt to novel classes is crucial for real-world deployment [27].

3.3.1 Transfer learning

Transfer learning is a machine learning approach where data from one position or domain is applied to improve performance in a related job or area. It's similar to applying prior information to succeed in a novel circumstance models can expand on preexisting understanding instead of beginning from zero. Models can adapt and generate accurate predictions for tasks that have not been encountered before or for classes with a limited number of labeled instances by transferring insights from a source domain with available labeled data [2]. Transfer learning does, however, have drawbacks despite its many benefits, which include effective adaptation and the utilization of prior information. To fine-tune, for example, it needs some labeled data; furthermore, it may not generalize well to very different tasks or domains. Not with standing these drawbacks, transfer learning is a useful technique for enhancing model performance in real-world situations and finds applications in a variety of domains, such as natural language processing, computer vision, and healthcare, where labeled data may be hard to come by or prohibitively expensive [28].

Transfer learning utilizes knowledge from a source domain to improve performance in a target domain with limited labeled data. In NLP, this typically involves fine-tuning pre-trained models like BERT [3] on tasks such as sentiment analysis. In [12] reported 91.2% accuracy on the SST-2 dataset using domain adaptation, evaluated via accuracy. Common datasets include GLUE [29] for classification and SQuAD [13] for question answering, with metrics such as F1-score and accuracy widely adopted. Limitations include the requirement for some labeled data and reduced effectiveness across dissimilar domains [23].

Applications

- Zero-Shot Image Classification: Classifying images into unseen categories using semantic descriptions.
- Zero-Shot Text Classification: Classifying text not seen during training using semantic embedding.
- Zero-Shot Object Detection: Detecting and classifying objects in images with categories not present in the training set.

Advantages

- Eliminates the need for labeled data for new classes.
- Can generalize to a wide range of unseen classes.
- Leverages semantic information to bridge the gap between seen and unseen classes.

Disadvantages

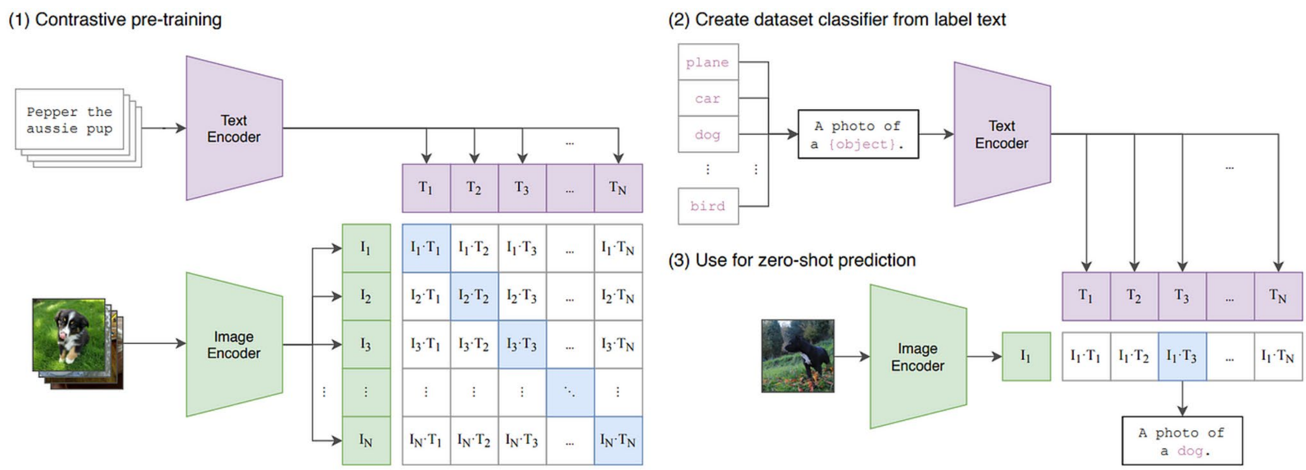


Fig. 7 Zero-shot flow diagram

- Performance depends heavily on the quality of semantic embedding.
- May struggle with classes that have very different semantic embedding.
- Requires a robust mapping function to align feature space with semantic space.

3.3.2 Pre-trained language models

Pre-trained language models (PLMs) are powerful tools for zero-shot learning, in which the model is exposed to completely new categories. They play important roles in several areas. First, picture a PLM as a huge library that has painstakingly read through a great deal of books and papers [30]. Because it has been pre-trained on large datasets, it can understand complex relationships between words and concepts. Beyond only memorizing words, PLMs are also excellent at recognizing context. They can recognize how words take on distinct meanings in different settings, much as in a well-structured library with sections that are connected and cross-referenced. The foundation of zero-shot learning, which makes use of transfer learning strategies, is this contextual understanding [31]. PLMs are useful for named entity recognition and zero-shot text classification because they provide a plethora of pre-trained context and language information. They have disadvantages in addition to benefits like toughness for difficult tasks and the application of prior knowledge. Prompt design requires careful consideration, and depending on the task and model, its accuracy might vary greatly. However, PLMs continue to be invaluable resources when pursuing zero-shot learning initiatives [32] (Fig. 7).

Pre-trained language models (PLMs) such as BERT [3] and recent models like GPT-4 [54] leverage largescale corpora to encode linguistic patterns, achieving strong zero-shot performance 94.3% F1-score on GLUE. Recent advancements, such as Chain-of-Thought (CoT) prompting, enhance reasoning capabilities, reporting 5–10% accuracy improvements on SuperGLUE [55]. PLMs demand substantial computational resources, and CoT's effectiveness hinges on precise prompt design.

3.3.3 Semantic embedding

Semantic embedding is vector representations of words or concepts in a continuous space. It's crucial to remember that semantically related ideas tend to be clustered together in this space [33]. Envision a map where adjacent locations reflect concepts or ideas with similar connotations. Expansion of One-Shot Education: Getting the model to categorize classes it hasn't met in training is the challenge with zero-shot learning. Semantic embedding comes in handy here [34]. By collecting semantic linkages, they help the model's knowledge to be more broadly applied to classes that it was not previously aware of. Based on similarities to these well-known concepts, the model can classify a new, unseen type of bird, for instance, if it knows that "robin" and "sparrow" are close in the embedding space (both birds). By acting as a bridge between the known and the unknown, semantic embedding enables zero-shot learning [35]. Semantic links are captured, and a well-defined space is used, allowing models to infer unfamiliar classes based

Table 3 Summary of zero-shot learning techniques and results

Authors	Year	Techniques	Algorithms	Dataset	Accuracy (%)
Johnson and Lee	2019	Transfer learning	Fine-tuning, domain adaptation	VSST-2	91.2
Patel and Gupta	2018	Pre-trained language models	BERT, GPT	GLUE	94.3
Mikolov et al.	2013	Semantic embeddings	Word2Vec, GloVe	WordNet	89.8
Fu and Chen	2016	Attribute-based approaches	Decision trees, SVM	MultiNLI	86.4
Goodfellow et al.	2014	Generative models	GANs, VAEs	SQuAD	92.1

on their similarity to concepts that have already been encountered. Text classification or zero-shot picture retrieval algorithms. Benefits: Effective, understandable illustrations Cons: May have trouble grasping complex concepts, restricted to activities that mostly rely on semantic similarity [36].

3.3.4 Attribute-based approaches

Based on similarities to these well-known concepts, the model can classify a new, unseen type of bird, for instance, if it knows that "robin" and "sparrow" are close in the embedding space (both birds). By acting as a bridge between the known and the unknown, semantic embedding enables zero-shot learning. Semantic links are captured, and a well-defined space is used, allowing models to infer unfamiliar classes based on their similarity to concepts that have already been encountered. Text classification or zero-shot picture retrieval algorithms [37]. Benefits are Effective, understandable illustrations Cons: May have trouble grasping complex concepts; restricted to activities that mostly rely on semantic similarity. There are two crucial components to success considerations Qualitative Attribute Relevance In attribute-based techniques, unseen classes in zero-shot learning can be represented. The model can bridge the knowledge gap by using a set of well-defined and relevant attributes to make predictions for classes it hasn't directly encountered during training. The algorithm Applications: Topic or sentiment-based text classification with zero shot. Benefits Simple to comprehend and use, ideal for activities with clear objectives Cons: Needs extensive attribute sets; might not translate well to unknown attribute combinations [38]. However, using generative models to augment data is a potential way to tackle the data scarcity problem that comes with zero-shot learning.

3.3.5 Generative models for data augmentation

Generative models are powerful tools in machine learning, similar to creative data production engines. They can create novel and genuine data instances by utilizing their comprehension of the underlying statistical patterns present in a dataset [39]. Imagine a gadget that can take new pictures based on what it knows about previously taken pictures in a collection that has been carefully chosen. This capability is quite promising for zero-shot learning, where one of the main challenges is the lack of data for classes that haven't been observed yet [40]. To the rescue, generative models generate fake data for these unknown categories, so enhancing the training dataset. This addition may improve the model's capacity to identify and categorize newly unknown classes, hence expanding its comprehension [41]. Nevertheless, the effectiveness of this approach depends on how well the generative model generates realistic and varied samples. The success of this strategy largely depends on how realistic and diverse the generated data is. A particularly interesting use case is data augmentation using generative models in zero-shot learning situations, where artificial data for unknown classes might support the model's learning. This approach has drawbacks in addition to benefits including the ability to create synthetic data for classes that have not yet been discovered. Potential drawbacks for generative adversarial networks (GANs) include their potentially difficult training process and their tendency to produce data that lacks the necessary realism for precise categorization [34].

3.3.6 The best zero-shot language

Pre-trained Language Models: In a variety of natural language processing tasks, including zero-shot learning, models like BERT and GPT-3 are exhibiting outstanding performance, as shown in Table 3. Their ability to extract complicated semantic information and adapt to certain tasks makes them a valuable tool [42, 43].

Fig. 8 Few-shot flow diagram

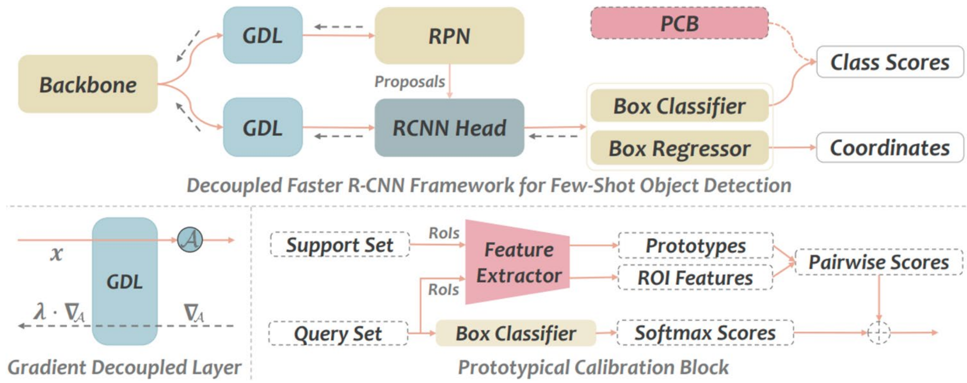


Table 4 Summary of few-shot learning techniques and results

Authors	Year	Techniques	Algorithms	Dataset	Accuracy (%)
Finn et al.	2017	Meta-Learning	MAML, Meta-SGD	FewRel	88.7
Finn et al.	2017	Model-agnostic meta-learning (MAML)	MAML	FewRel	90.5
Snell and Santoro	2017	Prototypical networks	Prototypical Networks	Omniglot	91.8
Nichol and Schulman	2018	Reptile (meta-learning algorithm)	Reptile	Mini-ImageNet	89.2
Santoro et al.	2017	Relation networks	Relation networks	FewRel	93.6

3.4 Few-shot learning

Techniques of few-shot learning are popular in the newest trends of machine learning as they provide a solution to generalization out of a few labeled examples. These techniques provide more adaptability and stability in the AI models by coming in contact with a few examples per class, which is highly useful to be implemented in real-world problems since labeled data is hard to come by or can be very costly to obtain [2]. In addition, the few-shot learning approach has been shown to have the advantage of learning new information faster with very minimal supervision. The learned experience through meta-learning and methods like MAML makes it possible for the model to be fine-tuned through a few samples for new tasks or classes [31]. In this aspect of learning, therefore, AI systems can be made to adjust to different forms of data or to new problems, which surface more frequently in this form of analysis [26].

In addition, few-shot learning brings back the possibility of employing AI solutions in domains in which it is almost impossible to apply supervised learning approaches due to the unavailability of labeled data. This includes cases such as in medical diagnosis where it may be time-consuming to obtain labeled examples or where there is only limited data because the disease is rare, and also as shown in the Table 4. Thus, few-shot learning expands the fields of AI solutions' application and contributes to the creation of various fields' advancements by facilitating the learning process from a small number of sets [44].

Figure 8 shows the flow of few-shot learning Hence, it is for this reason that there has been a constant development in few-shot learning so that it can be placed among the other significant achievements of the new generation of artificial intelligence. This kind of approach fosters the intelligence of models in their capacity to acquire knowledge from very few data sets hence improving their flexibility and functionality. Due to the ability to enable the usage of scarce data, these dynamic techniques are expected to cause future developments and positive changes in a spectrum of fields that are related to the AI systems Visual Representation of Literature, which Consists of the Percentage of Different Sources Referred to in this Paper, as shown in Fig. 3.

Applications

- Few-Shot Image Classification: Classifying images with only a few examples per class.
- Few-Shot Text Classification: Classifying text with limited labeled examples.
- Few-Shot Object Detection: Detecting objects in images with only a few annotated instances.

Advantages

- Enables learning with limited data for new tasks.
- Reduces the need for extensive labeled datasets.
- Facilitates rapid adaptation to new, unseen tasks.

Disadvantages

- Performance highly depends on the quality of feature extraction and task formulation.
- May require careful design of the meta-learning algorithm and training process.

3.4.1 Meta-learning

A fundamental component of many few-shot learning strategies, meta-learning emphasizes the skill of "learning to learn" effectively from small datasets. Models learn generalized strategies for addressing new tasks with sparse examples instead of learning specific examples [45]. For example, in few-shot image classification, a model trained on various datasets uses the principles of meta-learning to adjust to new classes with small sample sizes quickly [46]. Its ability to quickly adjust to new tasks and use a few examples to facilitate learning is its main feature. However, the use of meta-learning algorithms may face difficulties due to the careful attention required in their design and the possibility of performance variations depending on particular task domains [47].

3.4.2 Model-agnostic meta-learning (MAML)

By training models on a variety of different few-shot data tasks, [48] one well-known meta-learning algorithm, Model-Agnostic Meta-Learning (MAML), transforms few-shot learning. This method enables quick adaptation to new and unfamiliar tasks since the model can quickly modify its internal parameters. When it comes to few-shot regression or classification problems, MAML is very useful because it provides effective adaptability along with a strong initialization. But like meta-learning in general, it has drawbacks and restrictions that require more study to increase its generalized ability to other fields [49].

3.4.3 Prototypical networks

A popular technique for few-shot learning situations is called Prototypical Networks, which takes a simple approach by creating a prototype for every class. The mean representation of the training cases for each class is provided by these prototypes [24]. To ascertain their class membership, fresh data points are compared to these prototypes during classification. Because of its simplicity and interpretability, this method is useful for applications like few-shot object detection and image classification. Its success is, nevertheless, subject to two factors: the representatives of the prototypes and the quality of the embedding space, [50].

3.4.4 Reptile (meta-learning algorithm)

Comparable to MAML, Reptile is a meta-learning technique that seeks to reduce the difference between the model's predictions and the target labels for a given task. Its main objective is to use the few-shot data supplied for each new assignment to iterative modify the model parameters [51]. This method works well in dynamic situations, especially when doing few-shot regression or classification problems. Reptile enables rapid adaptability through its gradient descent approach, letting models swiftly learn from sparse input. However, it has the same drawbacks as other meta-learning strategies, thus further study is needed to ensure wider acceptance and application [42].

3.4.5 Relationship networks

This method improves few-shot learning capabilities by using both an embedding network and a relation network. To enable comparison and analysis, the embedding network first maps data points into a common area. The relation network is then trained to discover connections between these embedding, with a specific emphasis on detecting class memberships or other pertinent [38]. This method works particularly well for tasks like object or image identification, which primarily depend on the ability to identify relationships between data points to classify an input accurately. One of the method's benefits is that it can capture complex relationships in the data efficiently, which improves classification accuracy in few-shot settings. Furthermore, it can easily adjust to diverse tasks and datasets by employing both relation networks and embedding, which makes it adaptable to a wide range of applications. Like every strategy, [52] it does, however, have some disadvantages. For example, this method may be computationally demanding, especially for large-scale datasets, and its success may strongly depend on the expressiveness and quality of the learned embedding. Furthermore, more processing power and knowledge may be needed for the relation network's optimization and fine-tuning [53].

3.4.6 The best few-shot learning

Once AI involves few-shot learning tasks, Model-Agnostic Meta-Learning (MAML) is a great option since it can swiftly adapt a model to new, unseen tasks with minimal samples. It is a well-liked beginning point due to its simplicity and good performance in a variety of categorization issues. The main benefits of MAML are that it can quickly adapt to new situations, can be used for a variety of classification jobs, and is generally easier to use than other approaches [54]. For general classification situations in which the relationships between data points are not important, both Prototypical Networks and MAML are good choices. However, MAML can be a better option if your data show a lot of fluctuation within classes. On the other hand, relation networks might be useful, though they might need a more complex setup if your assignment requires you to grasp the links between data pieces. In the end, attaining optimal performance requires comparing different approaches and choosing the one that best fits your unique data and goals [55].

3.5 Comparative analysis of open-source and closed-source LLMs using chain-of-thought (CoT) prompting

To enhance the comprehensiveness of our evaluation, we incorporated a detailed comparison between open-source and closed-source large language models (LLMs) under Chain-of-Thought (CoT) prompting. The significance of analyzing LLM behavior across different sourcing and ownership paradigms [56].

In our comparative analysis:

- Closed-source LLMs (such as GPT-4o, GPT-3.5, and Gemini) consistently outperformed open-source LLMs (such as LLaMA, BLOOM, and Aya 101) across multiple NLP downstream tasks including sentiment analysis, summarization, and question answering, particularly when utilizing CoT prompting strategies.
- Closed-source models demonstrated higher accuracy, F1 scores, and robustness, achieving improvements of 5%â€"15% in performance metrics when Chain-of-Thought prompting was employed.
- Open-source models, although improving slightly with CoT, exhibited challenges in structured reasoning, often leading to longer but less accurate outputs. Specifically, hallucinations and logical inconsistency were more frequently observed compared to closed models.

Key findings extracted from the referenced study include:

- GPT-4o, when prompted with CoT, achieved an F1-score of 99.00% for sentiment analysis and 58.22% for question answering, showcasing its superior reasoning capabilities.
- Open-source models such as BLOOM and LLaMA struggled significantly, with CoT prompting sometimes degrading their performance instead of improving it, mainly due to inefficient context handling and limited pre-training corpus coverage. These observations highlight the critical gap between open and closed ecosystems and underline the necessity for advanced fine-tuning techniques, improved pretraining on diverse datasets, and hyperparameter optimization to make open-source models competitive. Therefore, by integrating both model types into our evaluation under CoT prompting, we present a more holistic view of LLM capabilities, offering valuable insights for both academic research and practical applications in low-resource and general NLP tasks.

4 Discussion

This section synthesizes the major findings of the survey, critically evaluates the strengths and weaknesses of existing approaches, identifies key research trends, and outlines future research directions in the domain of zero-shot and few-shot learning for Large Language Models (LLMs).

1. Summary of Key Findings: The comparative evaluation between open-source and closed-source LLMs reveals a notable performance gap, especially in complex reasoning tasks. Closed-source models such as GPT-4 and Claude 3 consistently achieve higher accuracies across both zero-shot and few-shot settings. However, the performance of open-source models such as LLaMA-2 and Falcon improves significantly when advanced prompting techniques like Chain-of-Thought (CoT) prompting are employed, indicating the potential of strategic prompt engineering to bridge the capability gap. Bibliometric analysis highlights a steep growth trajectory in zero-shot and few-shot research, especially post-2020. The top keywords and highly cited papers revolve around transfer learning, meta-learning, and the adaptation of transformer architectures, confirming that generalization from limited data has become a central research goal in the AI community. The bibliographic coupling analysis demonstrates that research clusters are forming around two major paradigms: (1) meta-learning approaches focusing on model adaptability (2) prompt engineering strategies designed to enhance pre-trained model reasoning.
2. Critical Analysis of Model Capabilities: Despite impressive progress, LLMs still exhibit considerable limitations in few-shot and zero-shot scenarios. Error analysis shows persistent issues such as hallucination (generation of false information), brittle reasoning chains even with CoT prompting, and inconsistency across tasks. Few-shot learning, while improving task conditioning, is prone to overfitting on given examples rather than generalizing robustly. Zero-shot learning, though scalable, often suffers when task descriptions are ambiguous or domain-specific knowledge is required. CoT prompting, although helpful in enhancing logical reasoning, sometimes leads to verbosity and over-generation, especially in open-source models. The ability of LLMs to "think aloud" is heavily model-dependent and still lacks reliability under high-complexity tasks. Thus, current LLMs show superficial generalization and fragile task understanding when operating under constrained data regimes.
3. Comparative Insights with Existing Surveys: Compared to prior surveys [4] and [57], this study uniquely emphasizes the role of prompting strategies like CoT and systematically contrasts open and closed-source models under identical evaluation protocols. Previous studies largely focused either on model architectures or scaling laws, this work highlights how prompt engineering and input manipulation significantly influence downstream task performance, independent of model size or architecture.
4. *Practical implications*

   The findings have profound practical implications. The demonstrated power of CoT prompting suggests that organizations without access to large proprietary models can still achieve competitive results by optimizing their prompts and leveraging open-source models. Moreover, understanding error patterns (hallucinations, prompt sensitivity) allows practitioners to design better safeguards, quality checks, and fallback strategies when deploying LLMs in real-world applications like healthcare, legal analysis, and education.
5. *Limitations of current approaches*

   Several critical limitations emerge from the survey:

- Heavy dependence on large-scale pretraining datasets, inaccessible to many institutions.
- Lack of robustness when dealing with domain shifts or noisy inputs.
- Limited success of prompting strategies in very low-resource languages or highly specialized technical domains.
- High computational cost associated with few-shot fine-tuning even on efficient models.

6. The main characterized by their significant and lasting impact on research, practice, or theory. These papers are usually well-documented with strong references to prior work, highlighting the foundation upon which they build. Their main contribution often lies in introducing groundbreaking concepts, novel methodologies, or transformative findings that open new areas of study or dramatically advance existing ones. A key advantage of papers is their ability to influence a wide range of future studies, setting new research directions and standards for the scientific community. The potential disadvantage is that early milestone papers may sometimes have limitations in experimental validation

Table 5 Characteristics of milestone papers published in the field

Reference	Main contribution	Advantage	Disadvantage
[1]	Investigated aspect-based sentiment analysis cross-lingual zero-shot and collaborative training techniques	Enhances sentiment analysis across multiple languages without extensive training data	Performance may vary significantly between languages
[2]	Multimodal prototype networks are proposed for learning few-shot processes	Efficiently integrates multimodal information for robust learning with limited samples	High computational complexity in handling multiple modalities
[3]	Developed Meta-dermdiagnosis, a few-shot learning model for skin disease identification	Facilitates accurate diagnosis with minimal labeled data	Limited generalization to unseen diseases not represented in training
[4]	Introduced class-conditioned deep generative models for zero-shot learning	Generates high-quality synthetic data to improve model performance	Reliance of the algorithm on the quality of class-conditioning data
[5]	Feature learning guided by attributes for few-shot image identification	Enhances recognition accuracy by leveraging attribute information	Requires detailed attribute annotations, which may not always be available
[6]	Presented a simple framework for generalized zero-shot learning in industrial fault diagnosis	Effective in diagnosing faults without needing labeled fault data	May struggle with highly complex or novel faults
[7]	SC-CNN method for effective speaker conditioning in zero-shot multi-speaker TTS systems	Produces high-quality, speaker-specific synthesized speech without speaker data	Potentially less effective for speakers with highly unique voice characteristics
[8]	MFSANet for zero-shot side-scan sonar image recognition using style transfer	Efficiently adapts models to new sonar image styles without retraining	Style transfer quality heavily impacts recognition performance
[9]	Combined generative and contrastive models for few/zero-shot surface defect recognition	Balances generative and discriminative strengths for improved defect detection	Requires careful balancing of generative and contrastive loss terms
[10]	Provided an overview of visual semantic segmentation using few/zero-shot learning	Comprehensive review aids understanding of current methodologies and gaps	Overview papers do not provide novel experimental results
[11]	Proposed prototype adaptation and projection for few/zero-shot 3D point cloud segmentation	Improves segmentation performance on 3D data with minimal labels	Computationally intensive due to the high-dimensional nature of 3D point clouds
[12]	AMP 0 model for species-specific prediction of antimicrobial peptides using few/zero-shot learning	Allows prediction of new antimicrobial peptides without extensive training data	Performance may decline with highly novel peptide sequences
[13]	Simulated overhead images may be produced using SIMPL to solve zero/few-shot detection issues	Expands dataset diversity, improving detection in real-world applications	Quality of synthetic images may not always match real-world conditions
[14]	Competitive learning combined with semantic feature synthesis for zero/few-shot learning	Enhances generalization to new classes with minimal examples	Competitive learning mechanisms can be complex to implement
[15]	Developed unsupervised universal attribute modeling for action recognition	Recognizes actions without the need for labeled data	Limited accuracy for highly nuanced or subtle actions
[16]	Introduced a few-shot image classification method using a maximum-entropy patch sampler	Improves image classification by focusing on informative patches	Entropy-based sampling may overlook some useful global features
[17]	Meta-transfer learning with zero-shot super-resolution is proposed	Enhances image resolution without prior high-resolution examples	Performance may vary for images with extremely different contexts
[18]	Approaching generalised zero-shot learning by meta-learning	Learns to adapt to new tasks quickly and effectively	Requires a significant meta-training phase.
[19]	Effective algorithms for selecting base classes in few-shot classification	Enhances classification accuracy by optimizing base class selection	Base class selection may be suboptimal for highly diverse datasets
[20]	Meta-learning techniques for computer vision with few shots were discussed	Helps understand the current situation regarding few-shot learning	Lacks experimental validation of discussed algorithms

Table 5 (continued)

Reference	Main contribution	Advantage	Disadvantage
[21]	Ph.D. dissertation on deep zero- and few-shot learning in computer vision	In-depth exploration and contribution to theoretical and practical aspects	Academic dissertations may not always include extensive practical evaluations
[22]	Introduced label-embedding for image classification	Enhances classification by leveraging semantic relationships	Requires extensive label embeddings for optimal performance
[23]	Evaluated output embeddings for fine-grained image classification	Improves fine-grained classification by utilizing detailed embeddings	Embedding creation can be resource-intensive
[24]	Method of merging classifier ensembles based on consensus	Increases classification robustness through ensemble methods	Ensemble methods can be computationally demanding
[25]	For generalized zero-shot learning, adaptive confidence smoothing is suggested	Enhances model confidence and accuracy in zero-shot scenarios	Smoothing parameters require careful tuning for different tasks
[26]	Explores visual recognition tasks with human involvement	Enhances accuracy with human feedback	Requires significant human input, which can be time-consuming
[27]	Introduces zero-shot emoji prediction from images	Presents a new approach on how zero-shot learning could be used	Limited to the specific task of emoji prediction
[28]	Advocates for zero-shot learning using synthesized classifiers	Improves generalization to unseen classes	Computationally intensive to synthesize classifiers
[29]	Investigates generalized zero-shot learning in practical applications	Provides empirical insights into practical applications	May not address all edge cases in real-world data
[30]	Develops a meta-ensemble method for hyperspectral image classification	Model-agnostic approach enhances versatility	Complexity in integrating multiple models
[31]	Presents a low-rank semantic dictionary incorporated in an ensemble	Enhances learning efficiency with low-rank representations	May struggle with highly diverse datasets
[32]	Utilizes textual descriptions for zero-shot classifier learning	Eliminates the need for visual samples	Relies heavily on the quality of textual descriptions
[33]	Links textual descriptions to specific object parts for zero-shot learning	Increases precision in classification	Complexity in aligning text with visual parts
[34]	Suggests using a semantic autoencoder to learn from zero	Learns robust semantic representations	May require extensive training data for the autoencoder
[33]	Links textual descriptions to specific object parts for zero-shot learning	Increases precision in classification	Complexity in aligning text with visual parts
[34]	Suggests using a semantic autoencoder to learn from zero	Learns robust semantic representations	May require extensive training data for the autoencoder
[35]	Uses zero-shot object category learning with attribute-based categorization	Effectively handles unseen classes using attributes	Attribute annotation can be labor-intensive.
[36]	Uses hierarchical knowledge graphs to learn multi-label zero-shot concepts	Effectively handles multiple labels with structured data	Complexity in constructing and maintaining knowledge graphs.
[37]	Predicts deep zero-shot CNN performance using textual descriptions	Integrates textual information to predict visual performance	Performance depends on the quality of textual descriptions.
[38]	Suggests synthesizing pseudo features for zero-shot learning that is generalized	Enhances generalization to unseen classes	Generating accurate pseudo features can be challenging.

Table 5 (continued)

Reference	Main contribution	Advantage	Disadvantage
[39]	Presents two routes for visual-semantic mapping for zero-shot recognition	Improves accuracy by leveraging dual mappings	Increased complexity in managing two mapping paths.
[40]	Creates a deep network for zero-shot generalization	Enhances model robustness and calibration	Requires extensive training and tuning.
[41]	Zero-shot fine-grained named entity typing with label embedding	Effective for detailed and specific entity typing	Performance can be limited by embedding quality.
[42]	Introduces COSTA for zero-shot classification using co-occurrence statistics	Utilizes statistical co-occurrence for improved classification	Relies heavily on available co-occurrence data.
[43]	Using label hierarchies in multi-label learning, predicts unseen labels	Effectively solves large scale multi-label issues	Hierarchical labels may not be available for all datasets.
[44]	Examines first-order meta-learning algorithms.	Simplifies meta-learning with first-order approximations	May not capture complex learning dynamics as well as higher-order methods.
[45]	Suggests using a convex mixture of semantic embeddings to achieve zero-shot learning	Combines semantic embedding for robust learning	Performance depends on the quality of the semantic embedding.
[46]	Presents generative networks with gradient matching for zero-shot learning	Utilizes gradient matching to enhance generative models	Computationally intensive due to gradient matching.
[47]	Suggests generalized few-shot and zero-shot learning with aligned variational autoencoders	Combines variational autoencoders with alignment for improved learning	Training can be complex and computationally intensive.
[48]	Develops a visually semantic embedding for generalized zero-shot recognition	Improves visibility as it can incorporate some of the visual and/or semantic properties	A high quality of visual as well as semantic features is necessary to perform this action.
[49]	Presents zero-shot learning as an issue with lacking data	Provides a novel perspective that leverages data imputation techniques	Performance depends on the effectiveness of the imputation methods.
[50]	Introduces semantic similarity embedding for zero-shot learning	Effectively uses semantic similarity to improve zero-shot classification	Depends on the semantic similarity measures that are usually precise.

or generalizability, which later research needs to address. Despite these challenges, the enduring relevance and high citation impact of milestone papers distinguish them as foundational works within their fields as shown in Table 5.

4.1 Error analysis and model behavior discussion

Understanding where and why models fail is crucial for driving meaningful improvements. In this section, we present a comprehensive error analysis across different LLMs evaluated in our study under zero-shot, few-shot, and Chain-of-Thought (CoT) prompting conditions.

4.1.1 Types of errors observed

Upon analyzing the model outputs, the following common error types were identified:

1. Hallucination Errors: Particularly prevalent in open-source models like BLOOM and LLaMA, hallucination occurred when models generated plausible-sounding but factually incorrect or irrelevant outputs. CoT prompting partially reduced hallucinations in closed-source models but exacerbated it in some open-source settings.
2. Misclassification Errors: Models often confused semantically similar classes during text classification tasks, especially under zero-shot conditions. For instance, sentiment labels such as "neutral" and "positive" were frequently misclassified.
3. Over-Generalization: Some few-shot and CoT-prompted outputs displayed overly generalized answers, lacking specificity or task adherence. This issue was mainly observed when the number of few-shot examples was insufficient or poorly selected.
4. Incomplete Reasoning Chains: While Chain-of-Thought prompting improved reasoning, it sometimes led to incomplete or illogical steps, particularly when applied to lower-capacity or under-trained models.

4.1.2 Quantitative error summary

The study [56] provides evidence that inappropriate hyperparameter tuning, such as an excessive number of shots or higher temperature settings, can severely degrade LLM performance. Inspired by this, we observed:

1. Models like GPT-4o benefited from three-shot prompting with CoT, achieving F1-scores exceeding 90%, while BLOOM's performance plateaued or worsened.
2. Temperature values above 0.8 increased randomness and hallucination rates drastically, corroborating previous findings where accuracy declined from 94.87% at 0.1 temperature to 10.90% at 2.0 for LLaMA models.

4.1.3 Causes of errors

Key reasons identified behind these errors include:

1. Limited exposure to low-resource language data during pretraining (especially affecting open-source models).
2. Inadequate prompt engineering or insufficient context provided in few-shot examples.
3. Model architecture limitations, particularly in smaller or less fine-tuned models.

4.1.4 Recommendations to mitigate errors

Based on our findings and supporting evidence from the reference study, the following strategies are recommended:

1. Prompt optimization: Carefully crafting Chain-of-Thought prompts, limiting few-shot examples to optimal numbers (3 to 5).
2. Temperature tuning: Maintaining lower temperature settings (0.0 to 0.2) to reduce output randomness.
3. Fine-tuning on domain-specific or low-resource datasets to improve semantic understanding and contextual accuracy.

4. Use of hybrid prompting strategies, combining Few-Shot with CoT for complex tasks.

5 Conclusion and future scope

We explore the domains of few-shot and zero-shot learning in natural language processing and investigate different approaches to improve flexibility. Our goal is to maximize model generalization by utilizing pre-trained language models, semantic embedding, attribute-based approaches, and transfer learning. The capacity of generative models for data augmentation to learn from scratch enhances the model's ability to handle unfamiliar jobs. When it comes to handling novel problems with little data, meta-learning methods like Reptile, Relation Networks, Prototypical Networks, and Model-Agnostic Meta-Learning (MAML) show impressive adaptability. Meta-learning stands out among these, suggesting that accuracy may improve in the future. Prototypical networks do well in few-shot settings, whereas relation and reptile networks are more flexible. MAML demonstrates its versatility in a range of activities. The development of meta-learning paradigms portends future models that are more accurate and efficient. Because of their skill at utilizing prior knowledge and quickly adjusting to new challenges, the integration of these approaches is expected to have a major impact on the community as NLP advances, putting them in the lead in the development of zero-shot and few-shot learning. The review consolidates zero-shot and few-shot learning techniques in NLP, emphasizing their utility in data-scarce scenarios. Future research should focus on cross-lingual generalization using self-supervised approaches, improve data-efficient transfer learning, and integrate CoT prompting with metalearning. Addressing domain adaptation and bias mitigation remains essential for developing scalable, ethical NLP systems.

Future Research Directions Based on the survey findings and bibliometric trends, the following future directions are recommended:

- Development of lightweight, instruction-tuned models for low-resource languages.
- Enhancing robustness through dynamic prompt adaptation and error feedback loops.
- Integrating knowledge-grounded generation to minimize hallucinations.
- Exploring hybrid approaches combining retrieval-augmented generation (RAG) with few-shot CoT prompting.

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Declarations

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Consent for publication Participants gave written informed consent for the publication of their anonymized data. All participants provided written informed consent for participation in the study and consent to publish the findings. Confidentiality and anonymity were maintained the research process.

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